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Building trust in generative AI

Essential evaluation techniques

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How Galileo and Google Cloud ensure GenAI quality and reliability

Generative AI (GenAI) holds immense promise, with results proving out as organizations move from experimental proofs of concept to full-scale production. However, ensuring these applications' quality, reliability, and safety demands that strong guardrails are put in place. The first step toward creating actionable evaluation metrics and trustworthy GenAI applications is having the right tools to evaluate these innovative apps across their development lifecycle.

This eBook will delve into the complexities of evaluating generative AI, offering practical insights and solutions to help you streamline the process and confidently deploy your AI applications. By employing evaluation-powered workflows, you can iterate more quickly, mitigate risks, and keep up with the rapid pace of change in this exciting landscape, setting your company up to unlock the true potential of GenAI.

Key takeaways include:

- Understanding GenAI's measurement problem
- The path to effective evaluation-powered workflows
- How Galileo and Google Cloud empower GenAI adoption and development

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Understanding GenAI's measurement problems

Generative AI's popularity is skyrocketing, with 65% of McKinsey survey respondents reporting that their organizations regularly use AI.[1] The rapid rise of generative AI has led to a dynamic and evolving market landscape that demands businesses adapt quickly to stay ahead. Part of that market evolution has sprung from complexity. AI applications are moving from simple, single-call large language models (LLMs) to more complicated retrieval-augmented generation (RAG)-based systems and multi-step agentic workflows that call on a mixture of functions and tools.

While exciting, this new generation of AI applications introduces new points of failure and an increased risk of harmful incidents like hallucinations and personally identifiable information (PII) leakage. Their behavior varies depending on input complexity, performance degradations can occur subtly over time, and their success criteria are often multi-dimensional. And in layered workflows like agentic RAG systems, finding what is causing an error among thousands of user queries and multiple steps of reasoning can be like looking for a needle in a haystack.

That increased risk is no easy hurdle to overcome for most organizations, leading to slower iteration cycles or even entirely stalling progress. According to Deloitte, three of the top four things holding organizations back from developing and deploying GenAI tools and applications are risk, regulation, and governance issues. In fact, 68% of respondents said their organization has moved 30% or fewer of their GenAI experiments fully into production.[2] Without the right tools in place to mitigate these risks and evaluate AI solutions across their lifecycle, companies end up falling behind the innovation curve in the GenAI space.

In the face of risk, regulation, and governance issues,

68% of companies

have moved 30% or fewer of their GenAI experiments into full production

1 McKinsey, "The state of AI in early 2024: Gen AI adoption spikes and starts to generate value."
2 Deloitte, "Now decides next: Moving from potential to performance."

Current evaluation methods: Human vs. LLMs

Businesses need the right evaluation metrics in place throughout the AI development lifecycle in order to identify potential bottlenecks, mitigate risks, and optimize performance. Today, organizations tend to use one of two approaches (or a mix of the two) for evaluating GenAI applications: human-based evaluations and LLM-as-a-judge. Human-based evaluation, frequently seen as the gold standard, involves humans reviewing and assessing the outputs generated by AI models, often using subjective criteria and qualitative feedback to measure aspects like accuracy, fluency, and coherence. LLM-as-a-judge, on the other hand, involves prompting an LLM to assess the quality of your GenAI outputs, including those generated by other models or human annotators.

The two options for evaluation each has its own set of drawbacks. While providing valuable insights, traditional human-based evaluation can be prone to human biases and inconsistencies. As a recent research paper put it, "Human evaluation is necessary, but **annotators are not infallible and may be biased, leading to evaluations that are useful but imperfect proxies of the desired objective.**"[3] Humans tend to score more assertive outputs as more factually correct, regardless of their actual content. Human evaluation is also expensive and time-consuming at scale. It requires coordination with annotators, custom web interfaces, detailed annotation instructions, data analysis, and considerations for employing crowd workers.

Meanwhile, using LLMs to evaluate other LLMs certainly offers scalability advantages. But using LLM-as-a-judge as the sole evaluator of your AI applications often raises concerns about cost, accuracy, and unintended biases. For instance, a pairwise comparison model is a good fit for determining relative quality between outputs but struggles to provide absolute performance metrics and scale efficiently as the number of comparisons grows exponentially with the number of samples or models being evaluated.

Issues with human evaluation

- Impact of confounding factors
- Bias in preference scores
- Low coverage of factual errors
- Harmful feedback loops
- Resource intensiveness

Issues with LLM-as-a-judge evaluation

- Nepotism bias
- Authority bias
- Beauty bias
- Verbosity bias
- Positional bias
- Attention bias (for lengthy text)

3 "Human Feedback Is Not the Gold Standard," Tom Hosking, Phil Blunsom, Max Bartolo; Cornell University.

Evaluation challenges across the AI development lifecycle

No matter which evaluation method a business is using, businesses will face challenges when embedding evaluation into the different stages of the AI development lifecycle—from building to shipping to scaling. It's not enough to catch errors in development or monitor performance degradations on a dashboard.

To build truly scalable, accurate AI solutions that users can trust, organizations need to embed evaluation-powered workflows across the AI lifecycle to both quickly debug and proactively prevent errors from occurring across each of these three stages.

Evaluation challenges across the AI development lifecycle

Building
Pre-production experimentation

At this stage, teams usually face the following challenges:

- Their minimally viable product (MVP) lacks guardrails.
- Manual evaluations are prone to inconsistencies and bias.
- Human-based evaluations also make it hard to get reliable and objective assessments of model performance.

Shipping
Production

The MVP has been moved into production, with new challenges:

- Monitoring live traffic and quickly debugging is difficult thanks to the complexity of new GenAI applications.
- Getting a high-quality test set and ground-truth is difficult.
- Manual evaluation workflows can slow iterative testing and measurement, but LLM-as-a-judge evaluations are not as accurate.

Scaling
Increasing usage

The product is now live and starting to scale, creating new concerns:

- LLM-as-a-judge evaluation is getting very costly and potentially slowing down the product.
- There's no easy way to keep harmful outputs from reaching existing users.

The path to effective GenAI evaluation

How do you embed evaluation-powered workflows into each stage of AI development—build, ship, and scale—so your organization can stay at the cutting edge? It goes beyond point solutions, which can become overwhelming to manage and may not give organizations the full visibility and speed of iteration they need to keep up. Instead, businesses need to invest in an evaluation ecosystem. By investing in the right evaluation tools and strategies, organizations can accelerate their AI initiatives, reduce the risk of errors and biases, and foster trust in their AI applications.

An evaluation ecosystem for each stage of AI development

Building
Pre-production experimentation

At the development and experimentation stage, teams need an evaluation system that:

- Provides access to the right metrics (system, privacy, security, and hallucination) for each model type, including more complex agentic RAG systems
- Enables rapid experimentation (across different models, prompts, and parameters) to build in confidence before moving to production
- Combines LLM-as-a-judge with human validation to achieve both scalability and accuracy in evaluation

Shipping
Production

Once the product has been moved into production, AI teams need evaluation-powered workflows that:

- Empower real-time monitoring and debugging to quickly iterate based on user inputs and block harmful toxicity
- Enable programmatic test set curation, input prompt iteration, and retrieval strategy experimentation
- Remove bottlenecks created by manual evaluation and intervention and prevent accuracy issues like test set drift
- Pinpoint issues and better understand how they propagate through the layers of an AI system with LLM tracing

Scaling
Increasing usage

As the AI application scales, an evaluation system should:

- Be adaptable to the evolving landscape of generative AI, incorporating new research and best practices to ensure ongoing effectiveness
- Enable teams to optimize their metrics for cost, accuracy, and latency
- Create a comfortable trade-off among these three aspects so teams can scale their solution to more users without incurring prohibitive costs or slowing down their applications

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Harnessing evaluation intelligence

How Galileo and Google Cloud empower generative AI

The Galileo Evaluation Intelligence Platform, integrated with Google Cloud Vertex AI, provides a comprehensive solution for evaluating, monitoring, and optimizing GenAI models. By embedding rigorous evaluation-powered workflows into all levels of the AI development lifecycle, Galileo and Google Cloud enable rapid iteration, comprehensive testing, continuous monitoring, and secure deployment of AI applications at scale.

Galileo and Google Cloud support each stage of the AI lifecycle

Building
Pre-production experimentation

- Mitigate LLM hallucinations and bias.
- Reduce the risks associated with generative AI, such as hallucinations, toxicity, and PII leakage, by utilizing Galileo's advanced evaluation capabilities and Google Cloud's secure environment.

Shipping
Production

- Accelerate shipping AI applications and iteration cycles and stop deploying in the dark.
- Streamline the evaluation process in production with Galileo's real-time monitoring and rapid debugging, integrated with Google Cloud's infrastructure, to bring your generative AI applications to market faster and adapt quickly to changes in the marketplace.

Scaling
Increasing usage

- Optimize your models with Galileo's comprehensive evaluation metrics and feedback loops, leveraging Google Cloud's scalable resources to achieve peak performance and accuracy.
- Intercept prompts and harmful outputs in real time to build an AI firewall that protects users.

Across the entire lifecycle, embed research-backed metrics and evaluation foundation models that provide guardrails for security, privacy, hallucinations, and RAG content.

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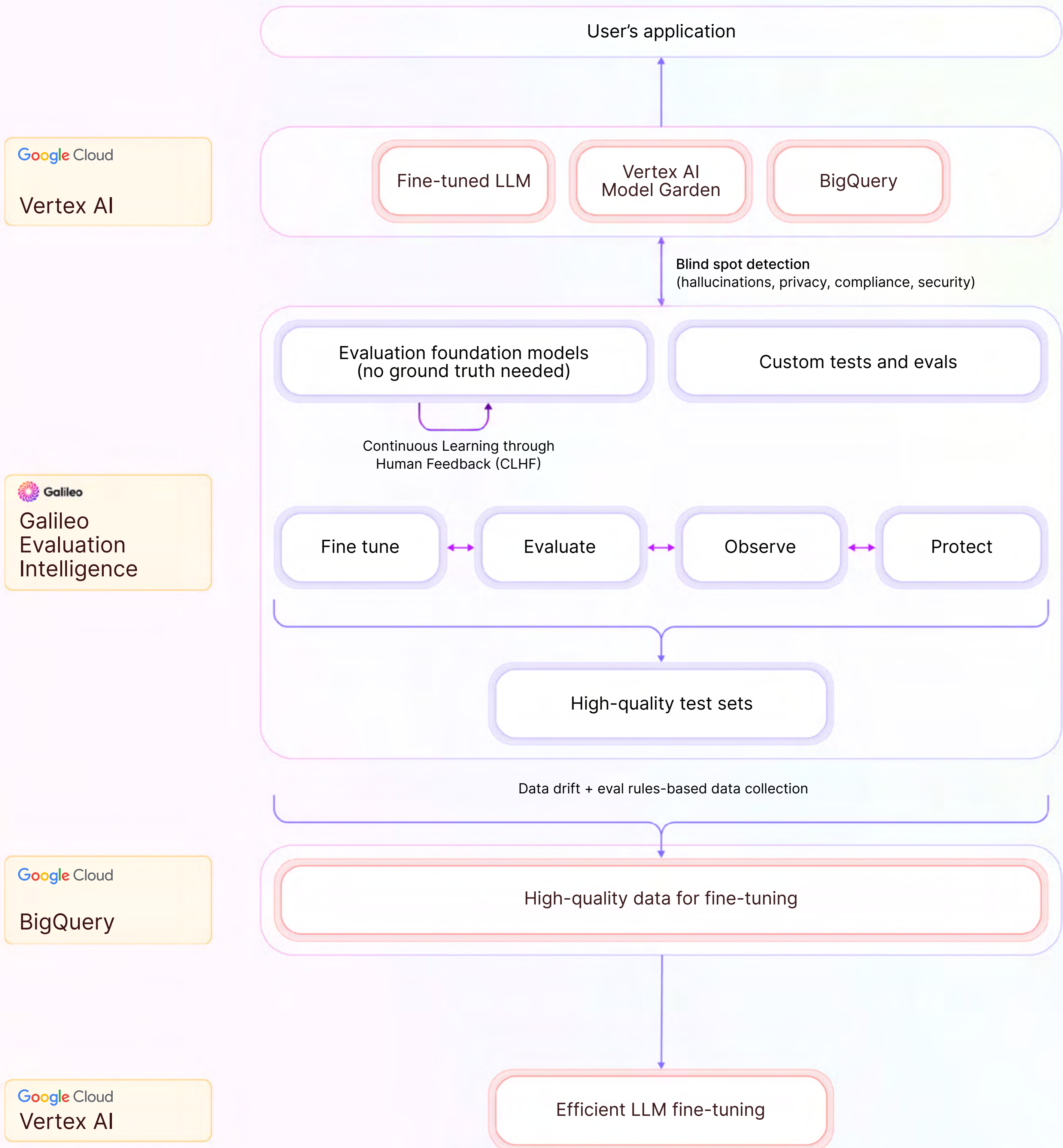
How Galileo and Google Cloud work together



Iterate faster with evaluation-powered workflows

Don't let the challenges of generative AI evaluation hold you back. Evaluation-powered workflows are the key to keeping pace with the rapid evolution of generative AI. By integrating continuous evaluation and monitoring into your development process, you can iterate quickly, identify and address issues in real time, and ensure your AI applications remain accurate, reliable, and aligned with the latest advancements in the field.

Embrace the power of the Galileo Evaluation Intelligence Platform and unlock the true potential of your AI applications. Visit Google Cloud Marketplace to learn more about how Galileo can help you build accurate, trustworthy, and reliable generative AI solutions.

Find us on the Google Cloud Marketplace

Let's go >